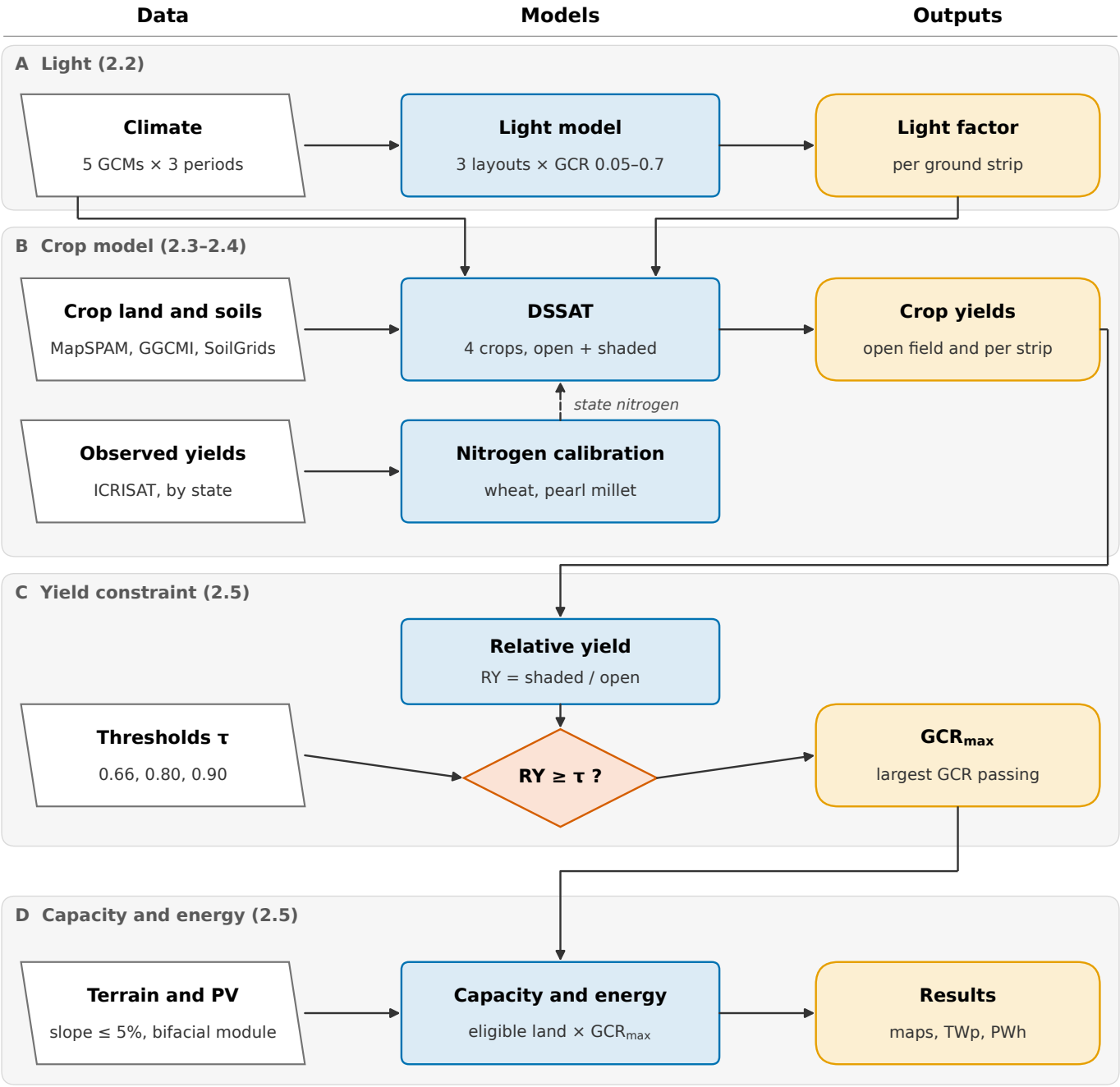
Data
Models
Outputs
A Light (2.2)
Climate
5 GCMs × 3 periods
Light model
3 layouts × GCR 0.05–0.7
Light factor
per ground strip
B Crop model (2.3–2.4)
Crop land and soils
MapSPAM, GGCMI, SoilGrids
DSSAT
4 crops, open + shaded
Crop yields
open field and per strip
state nitrogen
Observed yields
ICRISAT, by state
Nitrogen calibration
wheat, pearl millet
C Yield constraint (2.5)
Relative yield
RY = shaded / open
Thresholds τ
0.66, 0.80, 0.90
RY ≥ τ ?
GCRmax
largest GCR passing
D Capacity and energy (2.5)
Terrain and PV
slope ≤ 5%, bifacial module
Capacity and energy
eligible land × GCRmax
Results
maps, TWp, PWh